ATMOSYNC

Internship Mid-Review

MICRO-CLIMATE ARBITRAGE ANALYTICS

AtmoSync Analytics Engine & Data Pipeline

Progress Report: Week 1 & Week 2 Implementation

FOCUS
Containerized Climate Data

TECH STACK
Python • Pandas • GitHub

Status: Phase 1 Complete

PROJECT SCOPE & OBJECTIVES

Project Overview & Core Workflow

1. Requirement & Architecture

Studied AtmoSync functional requirements to construct containerized climate data workflows.

2. Data Cleaning & Analysis

Utilized Python for structural data cleaning, statistical validation, and climate profiling.

3. Climate Profiles & Next Steps

Evaluated micro-climates; Power BI dashboard integration scheduled as the upcoming phase.

Current Status: Python data pipeline, quality assurance, and statistical classification complete.

PHASE 1 DELIVERED

MILESTONE 1

Week 1 — Project Setup & Initial Workflow

Requirements & Environment Setup

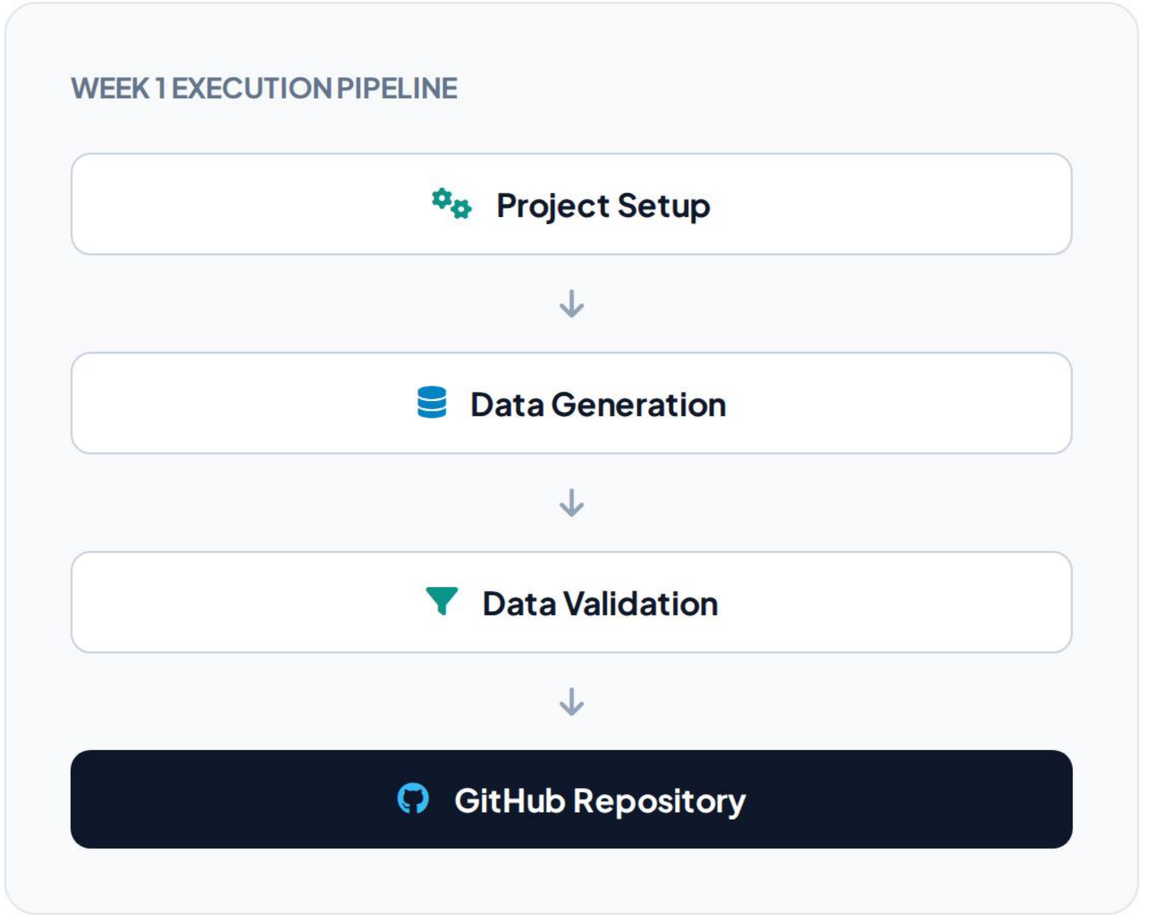
Studied AtmoSync specifications and configured local Python development environment.

GitHub Version Control Workflow

Initialized repository, branch structure, and commit protocols for seamless code tracking.

Initial Streaming Dataset Schema

Structured schema with Container ID, timestamp, temperature, humidity, and geo-coordinates.



MILESTONE 2

Week 2 — Final Dataset & Python Analysis

Dataset Generation

Synthesized 5,000 validated data points mapped across 100 distinct spatial containers.

- Cleaned & Null-Checked

Python Pattern Analysis

Evaluated thermal and humidity distributions using Pandas, NumPy, and Seaborn.

- Statistical Verification

Spatial & Climate Profiling

Aggregated container-level metrics to segment micro-climate profiles spatially.

- 4 Micro-Climate Clusters

Records Processed: **5,000 Rows** | Containers Tracked: **100 Units** | Key Output: **Micro-Climate Classification Model**

STATISTICAL HIGHLIGHTS

Key Analytics & Dataset Summary

5,000
Total Records

100
Containers Analyzed

26.37°C
Average Temperature

55.01%
Average Humidity

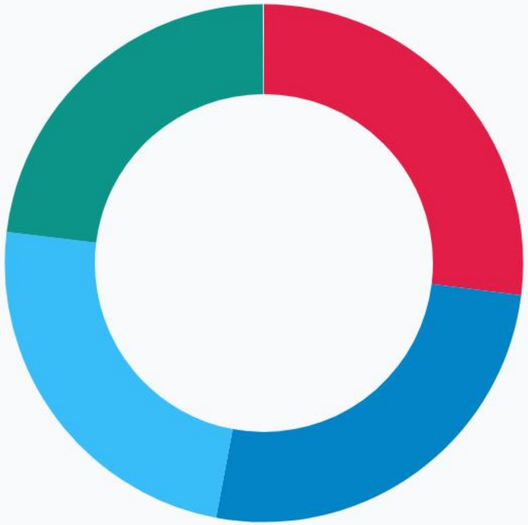
CORRELATION METRIC
Temp vs. Humidity Correlation **-0.47**

HOTTEST CONTAINER
CONT_0077 **42.46°C** avg

COLDEST CONTAINER
CONT_0082 **3.75°C** avg

MICRO-CLIMATE CLUSTERING

Climate Profile Analysis



Hot and Dry	27 Containers
Cool and Humid	26 Containers
Cool and Dry	24 Containers
Hot and Humid	23 Containers

Classification Logic: Containers were classified using average temperature and average humidity evaluated relative to dataset-wide averages.

ROADMAP & PROGRESSION

Implementation Timeline: Week 1 → Week 2 → Next Phase

WEEK 1

COMPLETED

- Project Setup
- Initial Dataset
- Basic Validation

WEEK 2

DELIVERED

- 5,000-Record Dataset
- Python Analytics
- Climate Classification
- Key Insights

NEXT PHASE

UPCOMING

- Power BI Dashboard
- Documentation
- Final Report
- Final Presentation / Demo



Thank You!

Questions & Discussion

AtmoSync: Micro-Climate Arbitrage Analytics Engine